# VoxMed

## Accessibility-First Medication & Allergen Assistant

EE471 Project Proposal

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# Outline

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# The Problem: Packaging Accessibility

- **Text-Heavy Packaging:** Food ingredients and medication instructions are printed in very small fonts, making them inaccessible to visually impaired and elderly users.
- **Hidden Hazards:** Important allergen warnings (e.g., peanuts, gluten, dairy) or specific contraindications are often buried in dense text.
- **Language Barriers:** Travelers and immigrants face difficulty reading product packages in foreign languages, risking accidental exposure to critical allergens.

# The Safety Risks

## Allergen Exposures

Accidental ingestion of allergens like peanuts or gluten can lead to severe health crises (e.g., anaphylaxis or severe celiac reactions).

## Medication Mismanagement

Misreading active ingredients or dosages poses high risks of drug-drug or food-drug interaction conflicts.

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# Introducing VoxMed

**VoxMed** is an AI-driven, voice-activated mobile assistant that acts as a real-time health and safety safeguard.

- **Instant Packaging OCR:** Captures food ingredients or medication labels and extracts all text automatically.
- **Smart Health Alignment:** Compares extracted ingredients with the user's allergy and medication profile.
- **Voice-First Interface:** Completely hands-free control via Speech-to-Text (STT) and Text-to-Speech (TTS).
- **Intelligent AI Reasoning:** Analyzes text using local Large Language Models (LLM) to determine safety.

# Interactive Features

- **Allergen Alert:** Checks for allergens (peanuts, milk, gluten, etc.) in food products.
- **Medication Advisor:** Reads out dosages, warnings, and warns about interactions.
- **Multilingual Support:** Automatically translates and reasons across different languages (e.g., scanning foreign packaging and warning in Turkish/English).
- **Follow-up Conversations:** Users can voice-ask questions: *"How much sugar is inside?"* or *"Does this cause drowsiness?"* and receive instant spoken answers.

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# Edge-Cloud Hybrid Architecture

**Edge & Local ML Server (Flask)**

- Runs locally on PC with GPU.
- **Whisper-tiny:** Local voice transcription.
- **SmolLM2-360M:** Fast local LLM reasoning.
- **OCR Engine:** Extracts text from captured images.

**Cloud Backend (AWS EC2 & S3)**

- Deployed in Docker containers.
- **Django Web Framework:** API gateway for profile management.
- **AWS S3 Storage:** Saves captured images.
- **Database (SQLite/PostgreSQL):** User health profiles & scan logs.

# DevOps & EE471 Concept Alignment

- **Containerization:** The cloud backend runs inside Docker containers, ensuring environment consistency.
- **CI/CD Pipeline:** Automated GitHub Actions workflow builds Docker images, runs lint/testing, and deploys directly to the AWS VM.
- **Hybrid Concept:** Low-latency heavy AI runs on the Edge (Flask local GPU), while lightweight persistent metadata is managed in the Cloud (AWS EC2 Django).

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# Development Timeline

- **Week 1 (June 10 - June 17): AWS & Cloud Setup**
  - Provision AWS EC2 instance.
  - Develop Django DB schemas for health profiles and audit logs.
  - Integrate AWS S3 storage API for image archiving.
- **Week 2 (June 17 - June 24): Local ML Server & Prompt Engineering**
  - Setup Flask local GPU server.
  - Integrate OCR engine and local LLM (SmolLM2).
  - Fine-tune prompts for allergen and drug interaction checking.
- **Week 3 (June 24 - June 30): Flutter App & Final CI/CD**
  - Design Flutter UI with accessibility options (Voice output/input).
  - Implement local Flutter Text-to-Speech (TTS).
  - Setup final GitHub Actions CI/CD pipeline for automated AWS deployment.

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# Expected Impact

- **Empowerment:** Enables elderly and visually impaired users to safely select and consume food or medicines independently.
- **Zero Latency AI:** Proves that heavy ML tasks (OCR, LLM, Whisper) can run efficiently on the Edge (local GPU) without requiring expensive Cloud GPUs.
- **Production-Ready DevOps:** Demonstrates full containerization (Docker) and professional cloud practices (AWS EC2/S3, CI/CD pipeline).

# Demo Walkthrough: Safe Chocolate Scan

## 1. User Health Profile

User Barış has a configured allergy profile on AWS: **Peanut (Yer Fıstığı) Allergy**.

## 2. Scan and Local Inference

Barış takes a photo of a chocolate bar. Local OCR detects: *"...Şeker, Kakao Yağı, Yer Fıstığı (%12)..."*. SmolLM2 cross-references and flags the danger.

## 3. Multi-Sensory Alert

Flutter app screen turns Red. Phone speaker speaks: *"Dikkat Barış! Alerjiniz olan yer fıstığı tespit edildi. Lütfen tüketmeyin."*

# Closing

Thank you for listening.

Any questions?